

Sandeep Sagar Madanu

Hyderabad, Telangana, India | +91-9515235158 | sandeepmadanu8@gmail.com
linkedin.com/in/madanusandeep-sagar | github.com/SandeepSagarMadanu | sandeep-portfolio.vercel.app

ABOUT

Data Scientist & AI/ML Engineer specializing in predictive modeling, deep learning, and LLM-powered applications. Proficient in end-to-end ML pipelines and currently researching explainable AI (XAI) and Vision Transformers for medical imaging.

TECHNICAL SKILLS

Programming & Querying: Python, SQL, Jupyter, Git/GitHub
ML / Deep Learning: Scikit-learn, TensorFlow, Keras, PyTorch, CNN, ViT, Feature Engineering, Hyperparameter Tuning, Cross-Validation
Data Science & Analytics: Pandas, NumPy, EDA, Statistical Analysis, Predictive Modeling, A/B Testing Concepts, KPI Reporting
NLP, LLMs & Prompting: LangChain, LangGraph, Hugging Face, LLaMA, GROQ API, RAG, Agentic AI, Prompt Design, Tool-Use Orchestration
Computer Vision & XAI: OpenCV, MediaPipe, Grad-CAM, SHAP, Radiomics, Image Classification
Visualization & Deployment: Matplotlib, Seaborn, Streamlit, FastAPI, REST APIs, Model Deployment
Domains: Medical Imaging AI, Autonomous Systems, Conversational AI, Predictive Analytics

EXPERIENCE

AI Researcher — Vision Transformers & Medical Imaging Dec 2025 – Present
Self-Employed (Independent Research) Hyderabad, India

- Directed investigation of hybrid CNN–Vision Transformer architectures for MRI-based MGMT prediction on 500+ scans, pursuing 1 Scopus-indexed publication in Medical AI / XAI.
- Constructed multi-modal ingestion frameworks unifying 3 input streams (MRI, radiomics, clinical records) via 5-fold CV, feature extraction, and reproducible benchmark suites.
- Benchmarked explainability with Grad-CAM and SHAP over 5 hold-out folds, assessing attention heatmap alignment with tumor boundaries for clinical reliability.

Data Science Intern Apr 2025 – Jun 2025
Brainovision Solutions India Pvt. Ltd. Hyderabad, India

- Produced 3+ Python ML workflows spanning ETL, model fitting, and performance audits on production datasets surpassing 10K rows across diverse industry sectors.
- Collaborated on 2 forecasting analytics engagements in 2-week Agile cycles, generating exploratory analyses and leader-board comparisons that lifted baseline macro-F1 by 8–12%.

Project Intern — AI & Autonomous Systems May 2024 – Sep 2024
Aigen Labs (Aeroforge Labs) Hyderabad, India

- Parsed sensor and telemetry streams from 2 UAV and autonomous-vehicle prototypes to fuel automation logic and vision-based proof-of-concepts.
- Built low-latency vision subsystems for robotic pick-and-place operations, sustaining 25+ FPS throughput on embedded unmanned platforms.

PROJECTS

Multimodal Image Query System — LLM & Prompt Engineering Python, GROQ API, LLaMA, FastAPI

- Launched a vision-language retrieval platform leveraging LLaMA via GROQ API, exposing 10+ FastAPI REST routes with iteratively refined prompt templates.

Agentic AI Chatbot with LangGraph Python, LangGraph, LangChain

- Created a conversational agent in LangGraph and LangChain, executing 5+ chained tool invocations per dialogue turn through persistent state graphs.

Medical Image Classification (Brain Tumor & Pneumonia Detection) Python, TensorFlow/Keras, CNN

- Fine-tuned 2 CNN classifiers on 3,000+ brain MRI and thoracic X-ray samples, reaching 92%+ hold-out accuracy and 0.89 macro-F1 for triage assistance.

Driver Drowsiness Detection & Traffic Sign Recognition Python, OpenCV, MediaPipe, CNN, Scikit-learn

- Shipped a driver-alertness tracker at 30 FPS via MediaPipe facial landmarks and a 43-class roadway sign detector at 96% test accuracy for ADAS applications.

EDUCATION

PGDM, Artificial Intelligence & Data Science (CGPA: 8.37) 2023 – 2025
ASB — Ashoka School of Business Hyderabad, India

B.Com in Commerce & Computer Applications (CGPA: 7.68) 2020 – 2023
Andhra Loyola College Vijayawada, India

CERTIFICATIONS

Python Using AI Workshop | Python Machine Learning 5-Day Bootcamp | AI and Data Scientist Certification